SANJAY BOLLINI

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EDUCATION

University of Washington Jun 2028
B.S. Computer Science | Minor: Applied Mathematics | GPA: 3.75 | Dean's List Seattle, WA
Coursework: Data Structures & Parallelism (CSE 332), Systems Programming (CSE 333), Computer Systems (CSE 351), Foundations of Computing (CSE 311), Software Tools (CSE 391), Numerical Linear Algebra
Certifications: Claude with the Anthropic API (Anthropic, Apr 2026), Multi-Agent Systems (Google), AWS Cloud Essentials, AI/Neuroscience (UW)

TECHNICAL SKILLS

Languages: Python, Java, C, TypeScript, JavaScript, SQL, HTML/CSS, Bash
Frameworks & Libraries: Next.js, Vue.js, FastAPI, Node.js, PyTorch, Pandas, NumPy
AI Tooling: Claude Code, Claude API, MCP (servers, tools, resources, prompts), multi-agent orchestration, LoRA fine-tuning, Ollama, prompt engineering
Systems & Infra: PostgreSQL, REST APIs, AWS, Git, GitHub Actions, circuit breakers, cron automation, x86-64 assembly, Vitest, Playwright, Zod

PROJECTS

AI Market Intelligence System 2025 - 2026
Python, PostgreSQL, Claude Haiku/Sonnet, Ollama (local LLM), yfinance, cron | github.com/Zing302/ai-market-intel
- Built a 4-agent pipeline (Analyst, Researcher, Trader, Manager) running 3x daily across 20 symbols in 5 sectors: SMA/RSI indicators, Haiku sentiment scoring, schema-enforced BUY/HOLD/SELL calls with conviction scores, and Sonnet escalation on high conviction or signal conflict.
- Production infrastructure: PostgreSQL schema, 7 cron jobs, per-source circuit breakers, 189 unit tests run on every push with GitHub Actions.
- 7-day outcome backfill scores each recommendation against realized price movement; a pluggable provider layer switches between Claude and local Ollama (Qwen3) with one environment variable.
- Solo-built and operated for 6+ months with zero manual intervention. Web dashboard in development.

BillPilot AI 2026
Next.js 15, TypeScript, Claude API, PostgreSQL | github.com/Zing302/billpilot-ai
- Built a medical billing assistant that benchmarks treatment costs, audits itemized bills for anomalies, and generates dispute-ready letters; recognized as a top project in the OpenAI x Handshake Codex Creator Challenge.
- Full-stack (REST API, React, PostgreSQL) with a pipeline from document ingestion to schema-enforced LLM output; tested with Zod, Vitest, and Playwright.

Quantum Password Generator 2025
Python, FastAPI, Qiskit, IBM Qiskit Aer | github.com/Zing302/quantum-password-generator
- Generates passwords from Hadamard-gate measurement entropy instead of a PRNG, on the Aer simulator or real IBM quantum hardware, with a FastAPI backend and an encrypted credential store.

Mindfulness4Success 2023
Vue.js, JavaScript | mindfulness4success.com
- Solo-built wellness web app launched at age 16, live since 2023; presented at the Youth Entrepreneur Summit.

EXPERIENCE

Cloud Girl Jun 2026 - Sep 2026
AI Developer Advocate Intern (with Priyanka Vergadia, Google Cloud Developer Advocate) *Remote*
- Designed and shipped a live MCP demo dashboard powered by 5 independent MCP servers (Calendar, Journal, Weather, Market & News, Log) with tools, resources, prompts, and a Claude agent tool-use loop; first demo delivered for the internship.
- Built a job-market analysis tool comparing AI Developer and Forward Deployed Engineer roles: a proxy-based scraper (Claude Code, DataImpulse) extracts required skills across postings, and a resume upload feature recommends the better-fit role. Filmed as the advanced demo in a tutorial.
- Prepared the demo and script for a 13:30 tutorial, "I Built Jarvis on a Free Colab GPU. It Talks Back.": LoRA fine-tuned Qwen3.5-4B on a free Colab T4 with Unsloth, built a ~300 example dataset, exported to GGUF, and deployed locally via Ollama with a local voice interface (faster-whisper, Kokoro). Zero cost, no data leaving the machine.

Alliance Endocrinology Feb 2020 - Present
IT Manager *Fort Worth, TX*
- Sole technical owner of all clinic IT since age 14, with no support team or escalation path: network infrastructure, Sophos firewall, IP cameras, and EHR systems (eClinicalWorks).
- Own HIPAA compliance across all clinical systems and data handling, device lifecycle, and access provisioning for clinical staff.
- Built the clinic website (HTML/CSS) and ran SEO and Google Ads that improved patient acquisition.

Mouri Tech Jul 2024 - May 2025
Data Engineering Intern *Irving, TX*
- Merged datasets, appended and deduplicated records, and added flag columns in SQL for analytics pipelines, covering schema design, query optimization, testing, and deployment.

RESEARCH

UT Dallas, Software Testing & Analysis Lab Jun - Aug 2024
ML & Formal Verification Research Intern *Dallas, TX*
- Integrated ML models into static analysis tools (JayHorn, JBMC, CBMC) to automate Java correctness verification across Ackermann, McCarthy 91, and Tower of Hanoi benchmarks; presented findings to faculty.

UW Machine Learning Research Club Sep 2025 - Present
Spiking Neural Networks Researcher *Seattle, WA*
- Collaborate on bio-inspired SNN architectures as energy-efficient, low-latency alternatives to transformer models, studying STDP and neuromorphic computing.